**KARNATAKA BOARD**

**CLASS 10**

**SCIENCE**

**2018**

**Max. Marks: 80**

**General instructions to the candidates:**

1. **This question paper consists of 42 objectives and subjective.**
2. **This question has been Sealed by reverse jacket .You have to cut the right side to open the paper at the time of commencement of examination.**
3. **Follow the instructions given against both the objective and subjective type of questions.**
4. **Figure in right hand margin indicate maximum marks of the question .The maximum time to answer paper is given at the top of question paper.**
5. **It ludes 15 minutes for reading the question paper.**

**For alternative are given for which of each of the following questions. Only one of them is correct or most appropriate alternative. Choose the correct alternative and write the complete answer along with its letter of alphabet.**

1. **"Coal is a non-renewable source of energy."**
   **(A) Because coal is replenished soon in nature.**
   **(B) Coal is abundant in nature.**
   **(C) The Reserve of coal are depleting at a faster rate and it is difficult to replenish.**
   **(D) Coal leaves residue when burnt.**

Ans: (C) The Reserve of coal are depleting at a faster rate and it is difficult to replenish.

2. **The living component of xylem tissue is.**
   **(A)Xylem vessel**
   **(B) Xylem parenchyma**
   **(C) Xylem tracheids**
   **(D)Xylem fibre**

Ans: (B) Xylem parenchyma

**3. Identify the type of amorphous silicon in the following.**

**(A)Does not burn in air**
**(B) Has dark grey colour**
**(C)Oxidizes at the surface level when heated in air**
**(D)Less reactive.**

Ans: (C) Oxidizes at the surface level when heated in air

**4. A man who is standing at a distance of 850 from a sound reflecting surface claps loudly if the velocity of the sound in air is 340 m/s ,then the time taken by the Eco to reach him is**

**(A) 5s**
**(B) 4 s**
**(C) 2.5 s**
**(D) 3s**

Ans: (A) 5s

**5. If the stages of human evolution is written in the descending order according to their cranial capacity then the correct order obtained is**

**(A) Homo habilis, Homo erectus, Homosapiens, Australopithecus**
**(B)Australopithecus, Homohabilis ,, Homo erectus ,HomoSapiens.**
**(C) HomoSapiens,Homo erectus,Australopithecus,Homo habilis.**
**(D)HomoSapiens, Homo erectus,Homohabilis,Australopithecus**

Ans: (D) HomoSapiens, Homo erectus, Homohabilis, Australopithecus

**6. Steam engine cannot be started instantaneously because**

**(A) The efficiency of the engine is low**
**(B)Steam should be produced by heating water**
**(C)The engine is bulky**
**(D) There is no spark plug.**

Ans: (B) Steam should be produced by heating water

**7. The principle of working of a motor is.**

**(A)There is a magnetic field around a current carrying conductor.**
**(B) When a magnetic field link which conductor changes, an induced EMF is generated in the conductor**
**(C)The change of current in one files EMF in a negative neighboring coil.**

**(D) A Conductor carrying electrical current experience mechanical force is kept in a magnetic field.**

Ans: (D) A Conductor carrying electrical current experience mechanical force is kept in a magnetic field.

**8. Antheridium of pteridophytes can be compared to**

**(A) Stamen of angiosperms**
**(B) Megasporophyll of gymnosperms.**
**(C)Carpels of angiosperms.**
**(D)Archegonium of bryophytes.**

Ans: (C) Carpels of angiosperms**.**

**9. The gas released when the sunlight breakdown Chlorofluoro carbon is**

**(A)Carbon dioxide**

**(B)Fluorine**

**(C) Carbon monoxide**

**(D)Chlorine**

Ans: (D) Chlorine

**10. The group of compounds when dissociate partially in aqueous solution is**

**(A)Hydrochloric acid, Nitric acid**

**(B) Carbonic acid, phosphoric acid**

**(C)Sodium chloride, acetic acid**

**(D)Copper sulphate, sugar solution**

Ans: (B) Carbonic acid, phosphoric acid

**11. The process related to organic compounds are given in column A and their procedures are given in column B. Match them and write the answers along with its letter:**

| Column A | Column B |
|---|---|
| (A) Preparation of Methane Gas | (i) Production of salts of fatty acids starting from oils or fats |
| (B)Substitution reaction | (ii) Conversion of liquid oils into solid saturated fats |
| (C)Hydrogenation | (iii) Heating fused sodium acetate with |

| | |
|---|---|
| | **sodalime** |
| **(D)Saponifiication** | **(iv) Heating an aqueous solution of ammonium cyanate** |
| | **(V)Burning of Methane in air** |
| | **(vi) Heating ethanol in presence of acidified potassium permanganate.** |
| | **(vii) Exposing the mixture of methane** |

Ans:

| Column A | Column B |
|---|---|
| (A) Preparation of Methane Gas | Heating fused Sodium Acetate with soda lime |
| (B)Substitution reaction | Exposing the mixture of methane and chlorine to ultraviolet light. |
| (C)Hydrogenation | Conversion of liquid oils into solid saturated fats Heating fused |
| (D) Saponification | production of salt of fatty acid starting from oils or fats |

**Answer the following questions:**

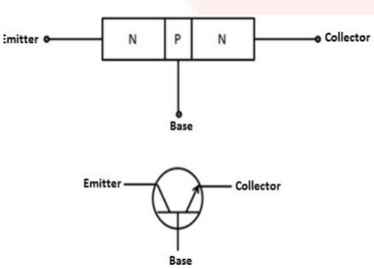
**12. Nowadays biodiesel is used in transportation vehicles as an alternate to diesel .write two advantages of this measure.**

Ans: The two advantages are:

It reduces the emission of harmful combustible gases.

It also helps in preserving diesel which is non -renewable source for long duration.

**13. Write the circuit symbol of p-n-p transistor.**



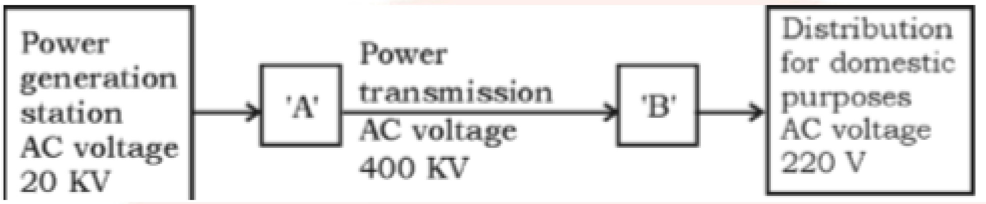

P-n-ptransitor

14. **Name the family and order of to which man belongs.**

Ans : Family - Primates

Order - Sapiens

15. **The schematic diagram indicating the transmission of electricity is given below:**




**Name the devices to be used in the places indicated as ‘A’ and ‘B’.**

Ans: ‘A’ Indicate Step Up transformer ‘B’ indicates step down transformer.

16. **How is silicon carbide prepared? Write one of its uses.**

Ans: Silicon carbide is prepared by heating silica with Coke in an electric furnace in the absence of air.

Uses - It is used in cutting and grinding of tools.

17. **In the manufacture of sugar the container of sugarcane juice is connected to a vacuum pump. Why?**

Ans: In the manufacture of sugar, the container of the sugarcane juice is connected to a vacuum pump because during the separation and drawing of crystals to remove water content present in the juices the juice is evaporated under reduced pressure. The water is let out in the form of steam through vacuum pump.

18. **If a person is having symptoms of thrust and frequent urination for a long time the blood capillaries in the retina of this person have ruptured causing blood entering into the vitreous humour making it Opaque name the eye disorder found in this person .**

Ans: The eye disorder found in this person is diabetes retinopathy.

**Answer the following questions: 16 ×2 = 32**

**19.”Manufacture of ethyl alcohol from molasses is good example for fermentation.”**

Ans: Fermentation is a process of chemical decomposition done by the microbial activity. Even in the manufacture of ethyl alcohol from molasses fermentation takes place by enzyme invertase and zymase which converts glucose into sucrose and this glucose into ethyl alcohol. It is a good example for fermentation.

**20. In animals breeding, write the two differences between outbreeding and hybridization**

Outbreeding

- -It is a conventional method.
- Superior male from a breed is crossed with the superior female of another breed.

Hybridization-

- It is a technological and hybridized method.
- Male and female breeds of two different species are crossed together

**21. What is Doppler Effect? Mention the two applications of droplet effect.**

Ans: An apparent change in the frequency of a wave when there is relative motion between the source and observer the Doppler Effect is produced .This phenomenon is known as Doppler Effect.

Two applications:

- Doppler Effect of light waves is used to track artificial satellites.
- Doppler effect of radio waves help in protecting the speed of vehicles to Radar gun

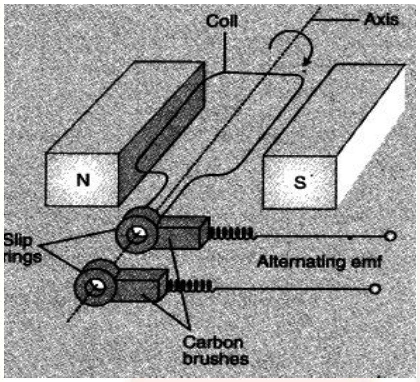
**OR**

**List the uses of ultrasonic waves due to a high frequency.**

**Ans:** Ultrasounds are used extensively in industries and for medical purposes. As mentioned above, Ultrasounds are high frequency waves (higher than 20 kHz). Even in the presence of obstacles, they are able to travel along well defined paths.

**22. Draw a diagram of AC dynamo and label the following parts:**

**(i)Armature (ii) brushes.**




**23. Observe the table in which size of different DNA fragments are given and answer the questions:**

| DNA fragments | A | B | C |
|---|---|---|---|
| Size (in Base Pairs) | 700 | 1500 | 3000 |

**(a) In the process of separating DNA fragments which fragment moves faster?**

Ans: (a) In the process of separating DNA fragments shorter element moves faster.

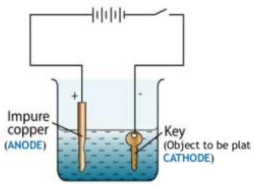
(b) After taking the samples of DNA from the enzymes which is introduced into a process called gel electrophoresis. In this process the fragments of DNA Break on the basis of net electric charge. The fragments break into longer and shorter segments and form a unique pattern. This pattern is similar to the finger print, they form bands of Unique pattern which is different from one person to another.

**(b) Explain the process of separating the DNA fragments.**

Ans: DNA fragments are separated as per molecular weight. Gel electrophoresis is the technique used to separate DNA fragments as per molecular weight.

Gel electrophoresis is basically the process by which we take the DNA, and run an electric charge through it. The DNA, being negatively charged by default, will move towards the positive side. As this happens, he DNA with lower density will travel less distance up. This can create a very unique pattern of DNA.

**24. Draw the diagram of the Apparatus used in electroplating and label the following parts:**




**Electroplating**

**25. What is monohybrid cross? Write the genotype ratio and phenotypic ratio of Mendel's monohybrid cross.**

Ans: The cross obtain by crossing the one pair of contrasting character is known as Monohybrid cross.

Genotypic Ratio - $1:2:1$

Phenotypic Ratio - $3:1$

**OR**

**Carl Correns conducted hybridization experiment using four o'clock plants. Draw the chequer board of F2 generation for the incomplete dominance phenomenon , when he crossed that a plant having red flowers(RR) with another plant with white flower (WW).mention its genotypic ratio**

Ans: Monohybird cross

| Gametes | R | W |
|---|---|---|
| R | RR | RW |
| W | RW | WW |

RR – Red flower

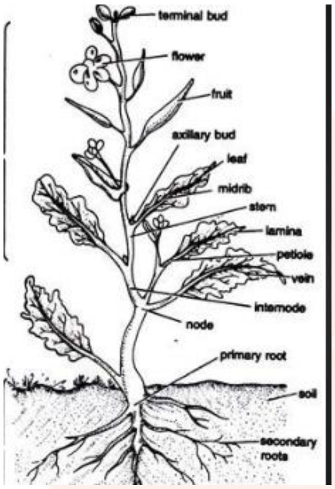
WW- White flower

RW-Pink Flower

Genotypictio- 1: 2: 1      Rr: Rw: Rw: ww

**26.Draw the diagram of a dicot plant and label the following parts .**

**(i) Flower (ii) Roots.**




**27. State Boyle's law .write the mathematical form of Boyle's law .Give an example of this law state.**

Ans: Boyle's law states that “At constant temperature the volume of a fixed mass of dry gas is inversely proportional to its pressure.”

Mathematical form

$V \alpha 1/p$

$V = K/P$

$PV = K$

Example when the balloon is breached due to pressure applied the balloon but this is an easy example for Boyle’s law.

**OR**

**State Graham's law of diffusion .Write the mathematical form of Graham law of diffusion. Give an example for this law**

Ans: Graham's Law which is popularly known as Graham's Law of Effusion, was formulated Thomas Graham in the year 1848. Thomas Graham experimented with the effusion process and discovered an important feature: gas molecules that are lighter will travel faster than the heavier gas molecules.

According to Graham's Law, at constant pressure and temperature, molecules or atoms with lower molecular mass will effuse faster than the higher molecular mass molecules or atoms. Thomas even found out the rate at which they escape through diffusion. In other words, it states that rate of effusion of gas is inversely proportional to the square root of its molecular mass. This formula is generally used while comparing the rates of two different gases at equal pressures and temperatures. Formula can be written as:

$$\frac{Rate_1}{Rate_2} = \sqrt{\frac{M_2}{M_1}}$$

$M_1$ is the molar mass of gas 1

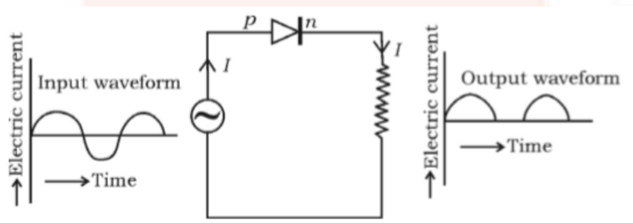
$M_2$ is the molar mass of gas 2

$Rate_1$ is the rate of effusion of the first gas

$Rate_2$ is the rate of effusion for the second gas

It states that rate of diffusion or effusion is inversely proportional to its molecular mass.

**28. Observe the following figure. Which property of diode is indicated here? Explain that property.**

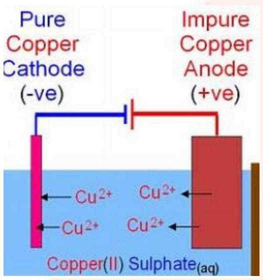



Ans: In the following figure, the ratifying property of diode of the p-region is connected to the positive terminal and if the, the n-region is connected to the negative terminal and when a strong source of AC current is allowed to pass the diode convert the AC current into DC current by rectifier in the AC. This property of diode is used in a device called rectifier.

**29. How is greenhouse effect caused? Explain name the greenhouse gases.**

Ans: Infrared radiations are released into the earth by the heating of solar radiation. This infrared rays are trapped by the atmospheric gases .As a result there will be increase in the earth's temperature. Especially it is trapped by the greenhouse gases. Thisphenomena is known as greenhouse effect. This is caused by the traffic of intraday that dress by the greenhouse gases .The greenhouse gases are Methane carbon dioxide and a small quantity of ozone gas.

**30. Draw the diagram of an electrolytic cell used in the purification of Copper and label the electrode having impure copper.**




**Purification of copper**

**31. Among the following identify the wrong statement with respect to a whale and write them correctly.**

**(i) A pair of lungs are respiratory organs**

**(ii) They do not have mammary glands.**

**(iii) Heart is four chambered.**

**(iv) They are oviparous**

Ans: In the following pair of for statement the statement (i) and (iii) are correct and two and four are wrong with respect to a whale.

(ii) Ans: They have mammary glands.

Ans: They are viviparous.

**OR**

**The organisms, (i) Amphioxus (ii) Balanoglossus, belong to which Phylum of chordata and why?**

**32. The molecular formula of the first member of a certain group of organic compound is $CH_2$ O. Determine the name and molecular formula of the third member of this group if the member of this group are in series what is the general name of this group of organic compounds?**

Ans: The molecular formula of the first member of a certain group of organic compound is $CH_2$ O The name and the molecular formula of the the third member of this group is propanol and its molecular formula is C3 $H_6O$. The members of this group are in homologous series .The general name for this group is called aldehydes.

**33. How is safety glass manufactured? Mention the use of safety glass.**

**OR**

**Name the types of paper having the following properties and mention one use of each.**

**(I) Porous and semi permeable**

Ans: Filter Paper.

It is used in the lab to separate solid residues from the liquid solution

**(II) Non sticky property.**

Ans: Wax paper.

It is used to wrap the food items such as cookies and ice-cream.

**34. The wavelength of a wave is 3 m. If the velocity of the wave is 330 m/s then find the frequency of that wave. Calculate the time period if the frequency of that wave is reduced to half its value.**

Frequency = 110 Hz.

Time period when frequency is half of its value

110/2 = 55 Hz

Time Period = 1/n Ans: Given: Wavelength = $\lambda$ = 3cm

Velocity = v 330 m/s

Frequency =?

Time period = s =?

$V = n\lambda$

$n = V/\lambda$

n = 330/3

= 1/55

=0.018 sec.

**Answer the following questions: 5× 3=15**

**35. Draw the diagram of a nuclear power reactor and label the following parts.**

**(i)The part that confines neutron to the core.**

**(ii) Radiation shield.**

**36. Explain the Havesian system of bone tissue.**

Ans: Structural and functional unit of bone tissue is Haverian system .The Haverian system is composed of dense matrix called ossien which is the rich in calcium and phosphate .the system consists of concentric circles called as lamellae. These lamella have small openings in them called as Lacunae. The Lamellae is connected to the adjacent lamellae by the thread like structure**.**

**OR**

**Explain the structure of cartilage tissue.**

Ans: Cartilage is a connective tissue consisting of a dense matrix of collagen fibres and elastic fibres embedded in a rubbery ground substance. The matrix is produced by cells called chondroblasts, which become embedded in the matrix as chondrocytes. That is, mature cartilage cells are called chondrocytes. They occur, either singly or in groups, within spaces called lacunae (sing. lacuna) in the matrix. The surface of most of the cartilage in the body is surrounded by a membrane of dense irregular connective tissue called perichondrium. This is important to remember especially because, unlike other connective tissues, cartilage contains no blood vessels or nerves except in the perichondrium. There are three different types (structures) of cartilage that have slightly different structures and functions. They are hyaline cartilage, fibrocartilage, and elastic cartilage.

**37. Explain the Intake stroke and compression Stroke in the working of petrol engine.**

Ans: Intake stroke: In this stroke the outlet valve is closed and the inlet valve is open. The mixture of air and petrol vapours is let in to the engine through inlet valve. The Piston is pushed downward.

Compressed Stroke- In this stroke both the walls are closed the Piston compressor is the year and petrol mixture towards the spark plugs

The Piston moves upward but the compressed temperature is not required to ignite the fuel.

**OR**

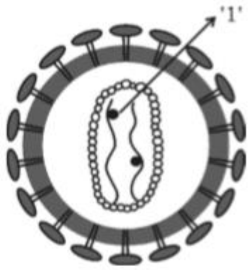
**Explain the working of diesel engine.**

**Ans:** Diesel engines burn fuel oils, which require less refining and are cheaper than higher-grade fuels such as petrol. During the combustion process, the stored chemical energy in the fuel is converted to thermal, or heat, energy. The temperature in each cylinder rises as high as 2,480 °C and creates pressures of about 100 kilograms per square centimetre. The pressure pushes against the tops of the pistons, forcing them to the other end of their cylinders. The pistons are connected by a rod or other suitable connecting mechanism to a crankshaft which they turn. In this way, a diesel engine supplies rotary power to drive vehicles and other kinds of machines.

In order for the compressed air inside the cylinders to ignite the fuel, it must have a certain temperature. The degree to which the temperature of the air rises depends on the amount of work done by the piston in compressing it. This work is measured in terms of the ratio between the volume of uncompressed air and the volume of the air after it is compressed. The compression ratio necessary to ignite the fuel depends on the size of the engine's cylinders. In large cylinders, the compression ratio is about 13 to 1. For small cylinders, it may be as high as 20 to 1. The average is 14.5 to 1.

Near the end of the piston's compression stroke, the fuel is injected into a cylinder. In order to have the fuel and air mix well, the fuel is injected under high pressure as a spray. Combustion usually starts just before the piston ends its compression stroke. The power of diesel engines can be increased by supercharging. This is the technique of forcing air under pressure into the cylinders

**38. Observe this figure and answer the questions given below:**




**(i) Name the part labelled as 1.**

Ans: The part Labeled as 1 is Reverse transcriptase enzyme.

**(ii)the hereditary material of the virus.**

Ans: The hereditary material of this virus is RNA (ribonucleic acid).

**(iii) The person infected by this virus is attacked by various diseases. Explain.**

Ans: The above virus is the HIV virus which causes the syndrome AIDS. This virus does not show the symptoms instant but reduces the immunity of the body gradually by various diseases like fever, cough etc. And WBC fails to identify the virus. This virus synthesize DNA and causes various diseases. Therefore the person infected by this virus is attacked by various diseases.

**39. The atomic numbers of five elements A,B, C, D and E are 6,, 8, 3, 7 and 9 respectively .**

**(i) Which is the element having highest electro positivity among these elements? Why?**

Ans: The element 'E' has the highest electro positivity among these elements because it loses one electron and attain octet structure. The element b is having leech metallic characters because it is a noble gas

**(ii) Which is the element having least metallic character among these elements? Why?**

Ans: The element 'B' is having least metallic characters because it is a noble gas.

**iii) What is your conclusion about the relationship between metallic character and electro positivity of an element?**

Ans: The relationship between metallic characters and electro positivity of an element is as we go across a period the electro positivity. Increases and metallic character decreases

**Answer the following questions: 3× 4 =12**

**40. a) Explain the red gaint stage of a star. What is the factor that decides the next stage after its red giant stage?**

Ans: After attaining the steady state the stars exert out more radiations and escape from the hold of gravitational pull.As a result of increase pressure and temperature the radiations are radiated out in a great speed .Consequently the outer part of the stars swell.After the losing all the radiation the star becomes red showing the radiation of radiation of low frequency .This stage is known as red giant stage of star.

The factor that decides the stage of star after its red giant is temperature and pressure of the star.

**(b) Define escape velocity with respect to earth. What do R and g indicate in the mathematical formula of escape velocity?**

Ans: The minimum velocity required for an object escape from the gravitational field of the earth is called and escape velocity.

$V_e=\sqrt{2Rg}$

In the mathematical formula of escape velocity 'R' indicate the radius of the earth and the 'g' indicates the gravitational constant that is acceleration due to gravity.

**OR**

**(a)Explain the supernova stage of a star. Mention the main feature of a black hole.**

**(b) State the new law of conservation of momentum."Propellants are necessary for the working of Rockets. Why?**

**41. (a) Observe the following chemical equations:**

**(i)** $Al_2O_3$ +NaOH→ 2 $NaAlO_2$ + $H_2O$

**(ii)** $Al_2O_3$ + 6HCl → 2 $AlCl_3$ + 3 $H_2O$

**What is the conclusion that you take about the nature of Aluminum oxide with the help of the equation. Give reason for your conclusion.**

Ans: (a) In the equation 1 and 2 we can take about the atmospheric nature of Aluminum oxide.

The reason for my conclusion is in both the equation. Aluminum oxide reacts with Sodium and chlorine and release the water molecule after reacting. Molten cryolite used as a solvent for molten aluminum reduces the melting point of aluminum and behave that electrolyte it is mixed with molten aluminum extraction of of graphite rods are used as cathode and carbon running is used as

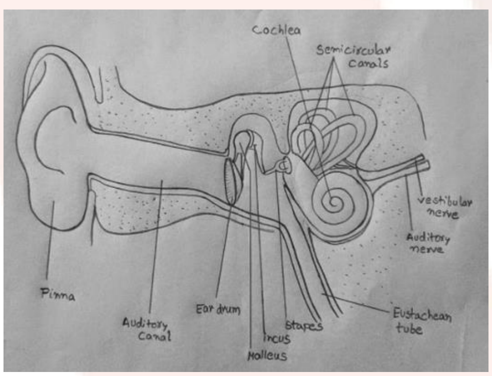
(b)Molten cryolite is mixed Alumina in the extraction of aluminum by .why? Name the substances that are used as anode and cathode in this method.

Molten cryolite used as a solvent for molten alumina. It reduces the melting point of alumina and behave that electrolyte. It is mixed with molten alumina in the extraction of aluminum.

Graphite rods are used as cathode and carbon running is used as anode.

**42. Draw the diagram showing the internal structure of human ear and label the following parts**

**(i) Malleus (ii) Auditory nerve**




**Human Ear**